

Introduction

I am a Grade 6 Science Teacher at school in Mumbai, teaching within the ICSE curriculum and serving as a Homeroom Teacher. My teaching philosophy is grounded in the belief that learning should be meaningful, inclusive, and connected to real life experiences. I aim to create a classroom where students feel safe to express ideas, ask questions, make mistakes, and participate actively in the learning process. Over time, I have realised that students engage more deeply when they are encouraged to think independently rather than simply recall information from textbooks. This belief has shaped my approach towards teaching science as an inquiry driven and discussion oriented subject. When I began reflecting on my classroom practice, I noticed a recurring pattern during questioning sessions. Although I regularly asked questions throughout lessons, the same small group of confident students consistently responded, while many others remained silent. Initially, I interpreted silence as lack of understanding or preparation. However, as I reflected more critically, I began to question whether my own questioning practices were unintentionally limiting participation and thinking. I realised that questioning was not merely about checking answers, but about deciding whose thinking becomes visible in the classroom.

This inquiry focuses on differentiated questioning as a strategy to support inclusion and deeper thinking in my Grade 6 Science classroom. Through this inquiry, I explored how structured and intentional questioning can create opportunities for all learners to participate meaningfully, regardless of their confidence level or language ability. The inquiry also investigates how questioning can move students beyond simple recall towards explanation, reasoning, and analysis. Although the Water Unit formed the primary context for this inquiry, the learning from this process has influenced my overall teaching practice across topics and lessons. This inquiry is also connected to wider educational priorities. The Programme for International Student Assessment highlights the importance of critical thinking, reasoning, and problem solving in modern education (OECD, 2019). Similarly, UNESCO (2020) emphasises the need for inclusive and equitable classrooms where all learners are able to engage meaningfully. These ideas reinforced my understanding that questioning is not simply a classroom routine, but a powerful pedagogical tool that shapes participation, inclusion, and thinking.

Context of Practice

This inquiry is situated within my Grade 6 Science classroom in Mumbai within the ICSE curriculum context. The classroom consists of learners with diverse academic abilities, confidence levels, learning preferences, and language proficiencies. While some students actively participated during discussions and responded confidently to questions, others hesitated to contribute, particularly during whole class questioning. Through observations and assessment analysis, I noticed that many students were comfortable recalling factual information but struggled when asked to explain concepts, justify answers, or apply understanding to unfamiliar situations. The ICSE curriculum places significant emphasis on conceptual understanding, application of knowledge, and interdisciplinary learning. However, despite these expectations, many classroom interactions remained surface level, with students focusing primarily on memorisation rather than deeper thinking.

Several key challenges emerged during classroom observations. These included unequal participation during questioning, varying levels of confidence, difficulty in expressing scientific ideas, and limited engagement with analytical thinking. Students with strong verbal communication skills often dominated discussions, while quieter learners remained passive observers. This created an imbalance in participation and raised concerns regarding inclusion within my classroom practice. The Water Unit was selected as the focus of this inquiry because it is both relatable and conceptually rich. Topics such as water sources, the water cycle, water pollution, and conservation provide opportunities for students to connect scientific understanding to everyday life experiences. At the same time, these topics encourage explanation, reasoning, and critical thinking. This context also presented practical challenges such as limited time for extended discussion, varying levels of vocabulary knowledge, and differences in learner confidence. These realities highlighted the importance of designing questioning strategies that are both cognitively challenging and accessible to all learners.

Another important contextual factor was the linguistic diversity within the classroom. Some learners were highly confident in spoken English, while others needed additional time to process scientific vocabulary before contributing verbally. This became particularly visible during higher order questioning tasks where students appeared to understand concepts but struggled to articulate responses using formal scientific language. Reflecting on this helped me recognise that silence in the classroom cannot always be interpreted as lack of understanding. In many cases, it reflected hesitation, fear of making mistakes, or difficulty expressing complex ideas publicly. This understanding challenged my earlier

assumptions about participation and encouraged me to think more carefully about inclusion. I realised that if questioning only privileges the fastest or most confident learners, then it risks excluding a large portion of the class from meaningful scientific dialogue. This insight became central to the direction of my inquiry.

Inquiry Focus

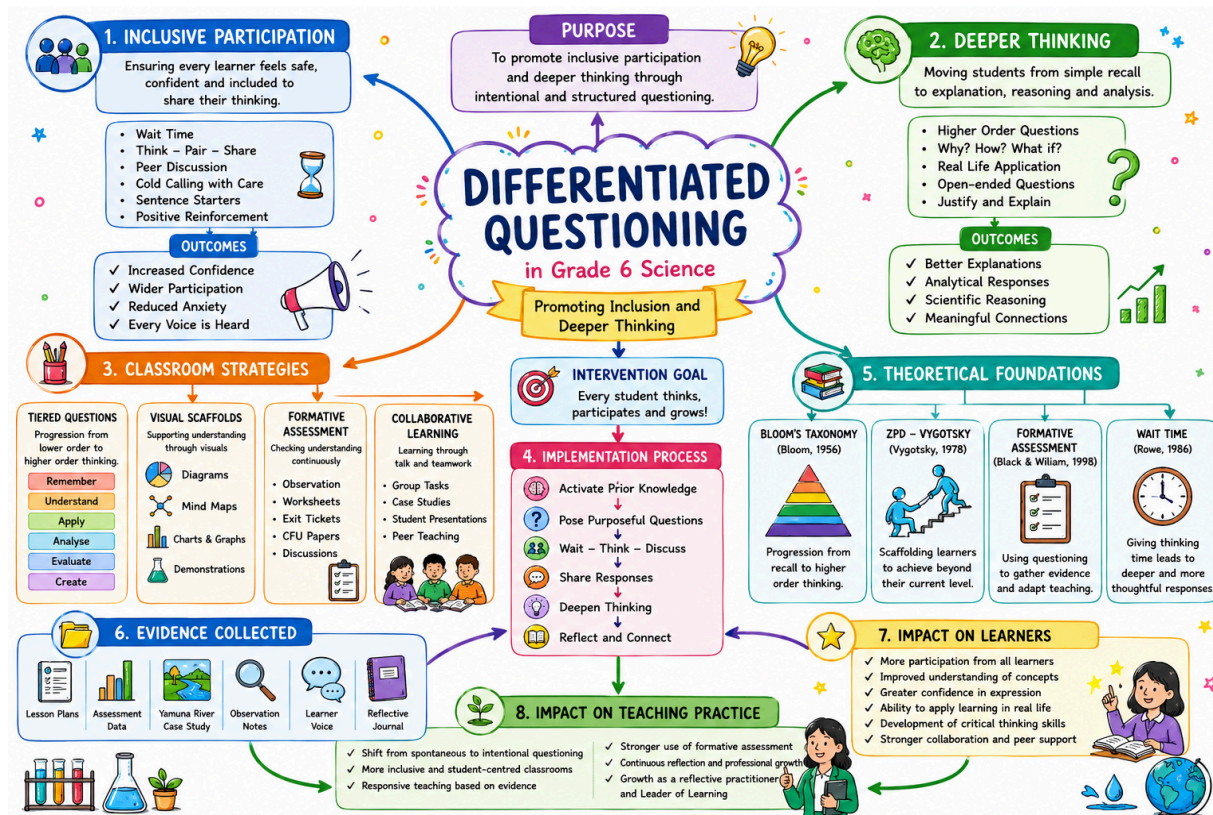
How can differentiated questioning support inclusive participation and promote deeper thinking in a Grade 6 Science classroom?

At the beginning of this inquiry, questioning in my classroom was largely spontaneous and unstructured. I often asked questions instinctively without fully considering their purpose or impact on learning. Through this inquiry, I began to understand differentiated questioning as a deliberate pedagogical strategy that influences both participation and thinking. Differentiated questioning refers to the intentional design and sequencing of questions at different levels of complexity to support diverse learners. In my classroom, questioning was structured to move from recall towards explanation, reasoning, and analysis. This approach aimed to ensure that all learners could access the discussion while also being challenged to think more deeply. This understanding is informed by Bloom's Taxonomy, which outlines the progression of cognitive skills from remembering and understanding to analysing and evaluating (Bloom, 1956). However, classroom experiences showed that students do not always move through these stages in a fixed sequence. Students often shifted between levels of thinking depending on the support, context, and confidence provided during discussions.

Vygotsky's concept of the Zone of Proximal Development also informed this inquiry by emphasising the role of scaffolding in supporting learners beyond their current level of understanding (Vygotsky, 1978). In my classroom, questioning became a form of scaffolding that supported students in expressing and developing their thinking. Questioning also functioned as formative assessment. Through classroom interactions, worksheets, and discussions, I was able to gather immediate evidence of student understanding and adapt instruction accordingly (Black and Wiliam, 1998).

Importantly, this inquiry positions differentiated questioning as serving two interconnected purposes. The first is inclusion, ensuring that all learners are able to participate meaningfully. The second is thinking, encouraging students to move beyond simple recall towards deeper conceptual understanding and reasoning. The inquiry was also influenced by reflective teaching frameworks discussed during the module. Schön's concept of reflection in action became particularly relevant during lessons where unexpected classroom situations required immediate adaptation (Schön, 1983). One example occurred during the water cycle lesson when technology failed and planned slides could not be displayed. Instead of stopping the lesson, questioning and pair discussion became the primary teaching strategy. Reflecting on this experience helped me realise that effective teaching depends less on resources and more on the teacher's ability to facilitate thinking and dialogue. This inquiry also aligns with dialogic teaching approaches which position classroom talk as central to learning (Mercer and Littleton, 2007). Through structured discussion and collaborative questioning, students were encouraged to construct understanding collectively rather than passively receive information.

Differentiated Questioning Intervention Framework: The below concept map 3.1 represents the differentiated questioning intervention used during the inquiry. It illustrates how questioning strategies such as wait time, scaffolding, think pair share, and tiered questioning were used to support inclusion, participation, and deeper thinking in the Grade 6 Science classroom. The framework connects theoretical perspectives including Bloom's Taxonomy, Vygotsky's Zone of Proximal Development, and formative assessment with classroom practice and learner outcomes. It also demonstrates my development as a reflective practitioner and emerging Leader of Learning through intentional and evidence informed teaching practices.



4. Portfolio of Professional Evidence

4.1 Lesson Plans and Scheme of Learning

The lesson plans developed for this inquiry integrated differentiated questioning throughout the teaching and learning process. Lessons focused on water sources, the water cycle, properties of water, and human impact on water resources. Each lesson included structured questions at different cognitive levels to encourage participation and thinking.

[Lesson Plan 1 link](#) , Lesson plan 1 extract:

Lesson Plan component	Annotation and reflection
Starter activity using real life images of water scarcity	This activated prior knowledge and encouraged emotional connection with the topic. Students related the discussion to water shortages in Mumbai.
Tiered questioning sequence	Questions moved from recall towards reasoning and analysis.
Think Pair Share strategy	Increased participation from quieter learners who hesitated during the whole class discussion.

This lesson plan focuses on developing students' understanding of different water sources through collaborative inquiry and structured questioning. It was selected as evidence because it demonstrates how differentiated questioning was used to engage all learners in exploring

a familiar yet conceptually important topic. The lesson begins with a real-life hook to activate prior knowledge, followed by group tasks where students investigate rainwater, surface water, groundwater, and glaciers. The use of guided questions supported learners in moving from identifying sources to analysing their usability and importance. This lesson provided insight into how collaborative learning, combined with purposeful questioning, can increase participation and deepen understanding. It also highlighted how students begin to make connections between availability of water and real-world challenges such as scarcity.

Lesson plan 2 Link

This lesson plan explores the stages of the water cycle using visual representation, discussion, and higher-order questioning. It was included as it reflects the shift from teacher-led explanation to student-centred inquiry. Through observation of diagrams and structured prompts, students were encouraged to explain processes such as evaporation, condensation, precipitation, and collection. A key feature of this lesson was the use of questioning to extend thinking beyond basic understanding, particularly when students were asked to identify other ways water vapour enters the atmosphere. This led to unexpected responses such as transpiration and respiration, demonstrating deeper conceptual thinking. The lesson provided valuable insight into how giving students time to think and discuss can enhance both engagement and the quality of responses, reinforcing the role of questioning as a tool for formative assessment.

This lesson provided valuable insight into how giving students time and space to think can improve both engagement and quality of responses. It also reinforced the role of questioning as formative assessment. [Lesson plan 3 Link](#)

This lesson plan focuses on the properties of water through demonstration and hands-on collaborative activities. It was selected because it highlights the effectiveness of experiential learning combined with differentiated questioning. Students engaged in practical tasks such as dissolving different substances to explore the concept of water as a universal solvent. The lesson also introduced complex ideas like anomalous behaviour through guided questioning and real-life connections. This approach enabled students to move from observation to

explanation and application. The lesson provided strong evidence of how active learning strategies increase engagement and support conceptual understanding. It also demonstrated that structured questioning can help students articulate their thinking more clearly while encouraging deeper exploration of scientific concepts.

Participation, collaboration and thinking rubric

Criteria	Emerging	Developing	secure	Advanced
Participation in Discussion	Rarely contributes	Responds when prompted	Participates independently	Leads discussion and supports peers
Explanation of Scientific Ideas	Gives one word answers	Gives simple explanations	Explains concepts clearly	Justifies ideas using evidence
Use of Scientific Vocabulary	Limited vocabulary	Uses some scientific terms	Uses scientific terminology confidently	Applies vocabulary accurately
Critical Thinking	Focuses on recall	Begins reasoning	Analyses ideas independently	Evaluates and creates solutions

This rubric was used informally during collaborative discussions and classroom observations. It helped track changes in participation,

confidence, and analytical thinking over time.

Reflection: The rubric also helped me become more aware of participation patterns within the classroom. I realised that some students who rarely contributed verbally during whole class questioning demonstrated strong understanding during pair discussions or written responses. This challenged my assumptions about participation and reinforced the importance of providing multiple ways for students to express understanding.

4.2 Assessment Data

Assessment evidence included worksheets, classwork, and a CFU (Check for Understanding) paper. These assessments were designed to reflect different levels of questioning. Analysis of student responses showed: Strong performance in recall-based questions, Improvement in explanation-based responses and emerging ability in analytical thinking This indicates that differentiated questioning supported learning progression, although further scaffolding is needed for higher-order responses.

Questions Responses 140 Settings

Total points: 13

140 responses

View in Sheets

Summary

Question

Individual

Insights

Average

10.69/13 points

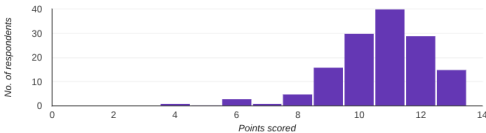
Median

11/13 points

Range

4-13 points

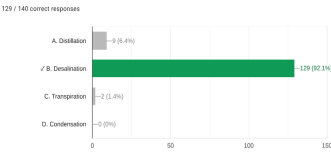
Total points distribution



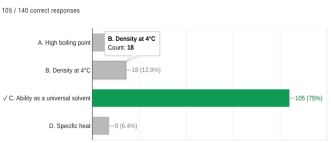
Questions Responses 140 Settings

Total points: 13

1. Which process is defined as the removal of dissolved salts from sea water?



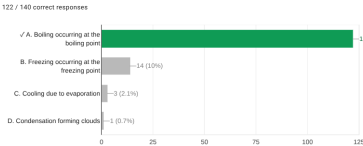
2. Which physical property allows water to transport nutrients in living organisms?



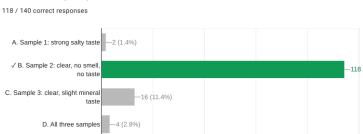
Questions Responses 140 Settings

Total points: 13

100°C, then remaining constant at 100°C. What does the flat part of the graph indicate?



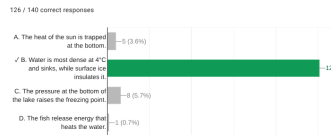
4. A class compares three water samples. Which sample best matches pure water at ordinary temperature?



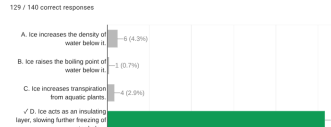
Questions Responses 140 Settings

Total points: 13

11. If the temperature in the village drops to -5°C, why does the water at the bottom of the local deep lake remain at 4°C?



12. Ice on the surface of a lake can help fish survive winter. How does this happen?



Q2. i. Aarav's body starts sweating because water regulates body temperature by producing sweat. This sweat prevents overheating cooling his body down. In this way, his body stays safe and healthy.

i. After running in the playground on a hot day, Aarav's body starts sweating. Explain how this helps his body stay safe and healthy. [1]

ii. After a long game, Aman drinks water along with his food. How does water help his body use the nutrients he has eaten? Water dissolves nutrients and carries them to all parts of the body. Through the bloodstream, these nutrients help in the important process of the body like protection. [1]

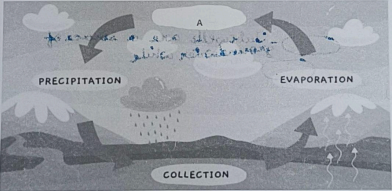
Q.3.A. Define the boiling point of water. *Boiling point is the temperature at which liquid starts to boil. It is called its boiling point. It is the temperature at which the vapour pressure of liquid becomes equal to its atmospheric pressure.*

Q.3.B. A family went camping in the mountains and noticed that their food was not cooking properly even after a long time. Explain why this happens. *The boiling point of water decreases with a decrease in pressure. This is because when pressure is acting on the surface of a liquid, the liquid can escape as water vapour easily. Due to this reason, water boils at a lower temperature in the hills. Hence, it takes longer to cook food in hilly regions.*

Q.3.C. Differentiate between the surface water and ground water based on the given criteria-purity and accessibility. [2]

Criteria	Surface water	Ground water
Purity	Can be impure due to waste and exposure to contaminants.	Tends to be pure as it is naturally filtered through layers of rock and sand.
Accessibility	More easily accessible.	Less accessible.

Q.4. Observe the below diagram and answer the following questions.



i. Identify the process labelled as A. The process A is condensation. [1]

ii. How do plants add water vapour to the atmosphere? *Plants add water vapour to the atmosphere through transpiration. Transpiration is the process where plants take in water through their roots and then release some of that water as vapour through tiny openings in their leaves. This water vapour goes into the atmosphere, contributing to the moisture in the air.*

Q.1.A. Choose the correct answers to the questions from the given options. (Do not copy the question, Write the correct answer only.)

i. Respiration produces carbon dioxide, water and _____. [1]

a) Heat
☒ b) Energy
 c) Oxygen
 d) Nitrogen

ii. Assertion (A): Groundwater is usually cleaner than river water. [1]
 Reason (R): Groundwater gets filtered through layers of rocks and sand.
☒ a) Both A and R are true, and R is the correct explanation of A
 b) Both A and R are true, but R is not the correct explanation of A
 c) A is true, but R is false
 d) A is false, but R is true

Q.1.B. Find the odd one out and give reason for the odd one out.

i. River – Lakes – Oceans – Tube wells *- Tube wells are a source of groundwater while all the others are examples of surface water.* [1]

ii. Drinking – Cooking – Bathing – Smelting *Smelting is used in industries while all the others are used for household needs.* [1]

Q.2. Read the passage below and answer the following questions:

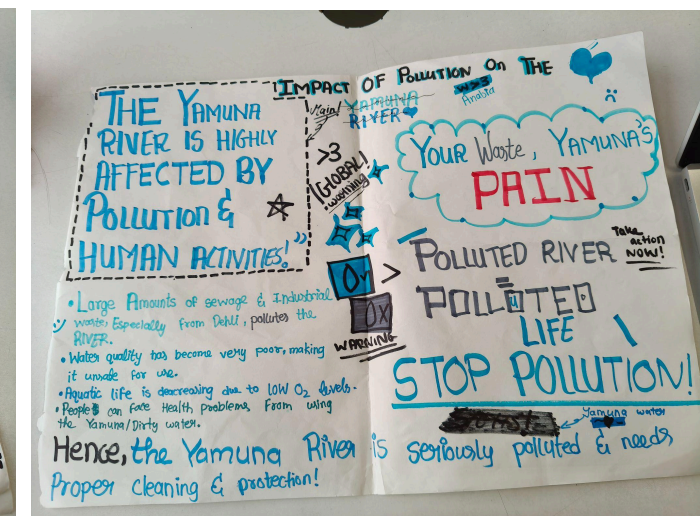
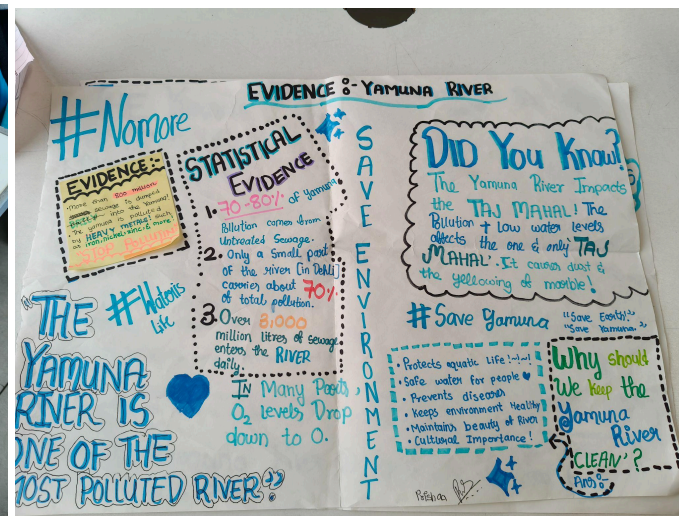
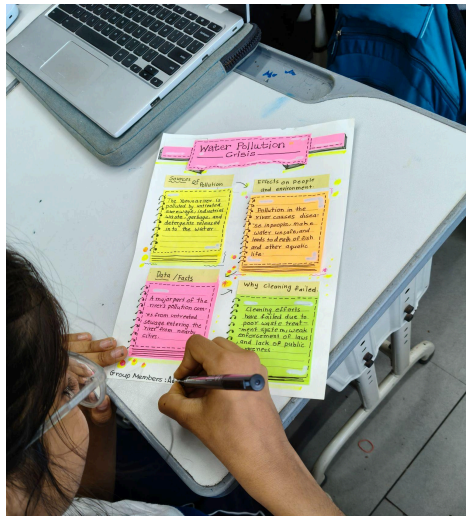
Water plays a very important role in the human body. It helps dissolve nutrients and carry them to all parts of the body through the bloodstream. Water also regulates body temperature by producing sweat and helps remove waste through urine, sweat and faeces. In addition, water protects sensitive parts of the body such as the spinal cord, joints and tissues. Since our body continuously loses water, we must drink enough water every day to stay healthy.

The inclusion of multiple-choice, reasoning-based, and application questions allowed for a range of responses. For example, identifying groundwater as cleaner demonstrates recall, while explaining the reasoning reflects conceptual understanding. Student responses indicate that most learners achieved the intended outcomes, particularly in identifying water sources and understanding basic concepts. However, extended reasoning remains an area for development. This artefact highlights how assessment can function as a continuation of differentiated questioning, providing valuable evidence to inform next steps in teaching.

4.3 Annotated Student Work

Student work samples provide direct evidence of learning. Worksheets on water usage showed that students could identify uses of water and provide basic explanations. Responses on climate change demonstrated an emerging understanding of cause and effect relationships. The Yamuna River case study poster stands out as a strong example of higher-order learning. Students identified sources of pollution, analysed impacts, and evaluated why solutions have failed. This reflects movement towards analysis and creation.

Evidence observed	Professional reflection
Students identified multiple pollution causes	Demonstrates movement beyond recall towards analysis
Students connected pollution to health effects	Shows application of scientific understanding
Group included awareness slogans	Reflects creativity and real world connection



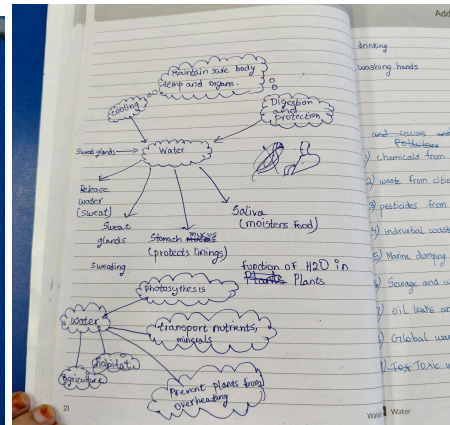
This activity demonstrated that students engage more deeply when science is connected to authentic environmental issues. Differentiated questioning supported students in moving from factual understanding towards evaluation and problem solving. This activity was also significant because it revealed the power of collaborative inquiry. Students challenged each other's ideas, negotiated meaning, and justified opinions using evidence gathered during research discussions. This demonstrated that questioning does not only come from the teacher, but can also emerge through peer dialogue.

5 Write whether the given statement is true or false. Correct and rewrite the false statements.

(a) Water makes up over 60% of the adult human body and does not serve a variety of purposes.
False
 Water makes up over 60% of the adult human body and serves a variety of purposes.

(b) Water allows waste products to dissolve and leave the body through sweat, faeces, and urine.
True

(c) Water does not regulate the temperature of our bodies.
False
 Water regulates the temperature of our bodies



This mind map illustrates the student's understanding of the functions of water in human body systems and plants, including digestion, temperature regulation, nutrient transport, and photosynthesis. The use of a visual representation shows the student's ability to organise and connect ideas meaningfully.

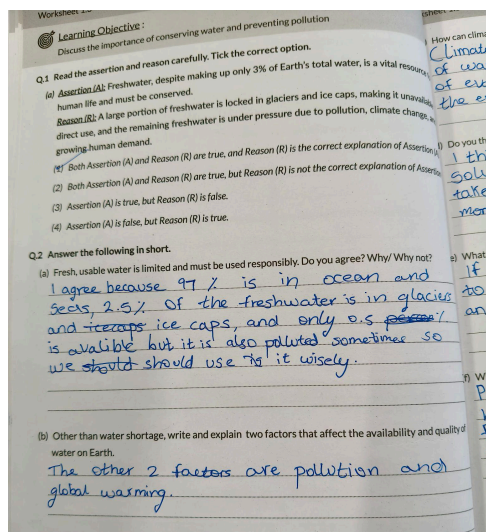
Additional Notes

Earth
 71% = water
 29% = land
 Freshwater
 2.6% = freshwater
 0.5% = available
 97% = salty
 dirty water

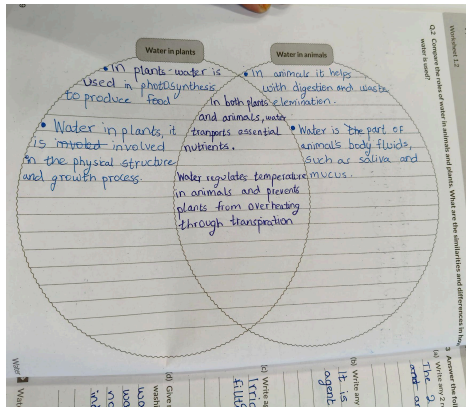
Drinking
 Washing hands
 and - oceans and
 Pollution
 1) chemicals from factories
 2) waste from cities
 3) pesticides from farms
 4) industrial waste
 5) home dumping
 6) sewage and wastewater
 7) oil leaks and spills
 8) global warming
 9) toxic waste

Impact on water
 1) Aquatic life
 2) water shortage
 3) loss of biodiversity
 4) spread of diseases

This student's notes capture key concepts related to global water distribution and causes of water pollution, including percentages of freshwater availability and multiple pollution sources such as industrial waste, sewage, and oil spills. The inclusion of both data representation (pie diagram) and extended lists indicates that the student was able to synthesise information gathered through group discussion and guided questioning.



This worksheet provides evidence of the student's understanding of water scarcity and the need for conservation. The student correctly identifies that only a small percentage of Earth's water is usable and attempts to justify responsible usage using numerical data (97% saltwater, 0.5% available freshwater). This reflects conceptual understanding developed through class discussions and questioning sequences. The assertion-reason question has been answered correctly, indicating the student's ability to engage with logical reasoning tasks.



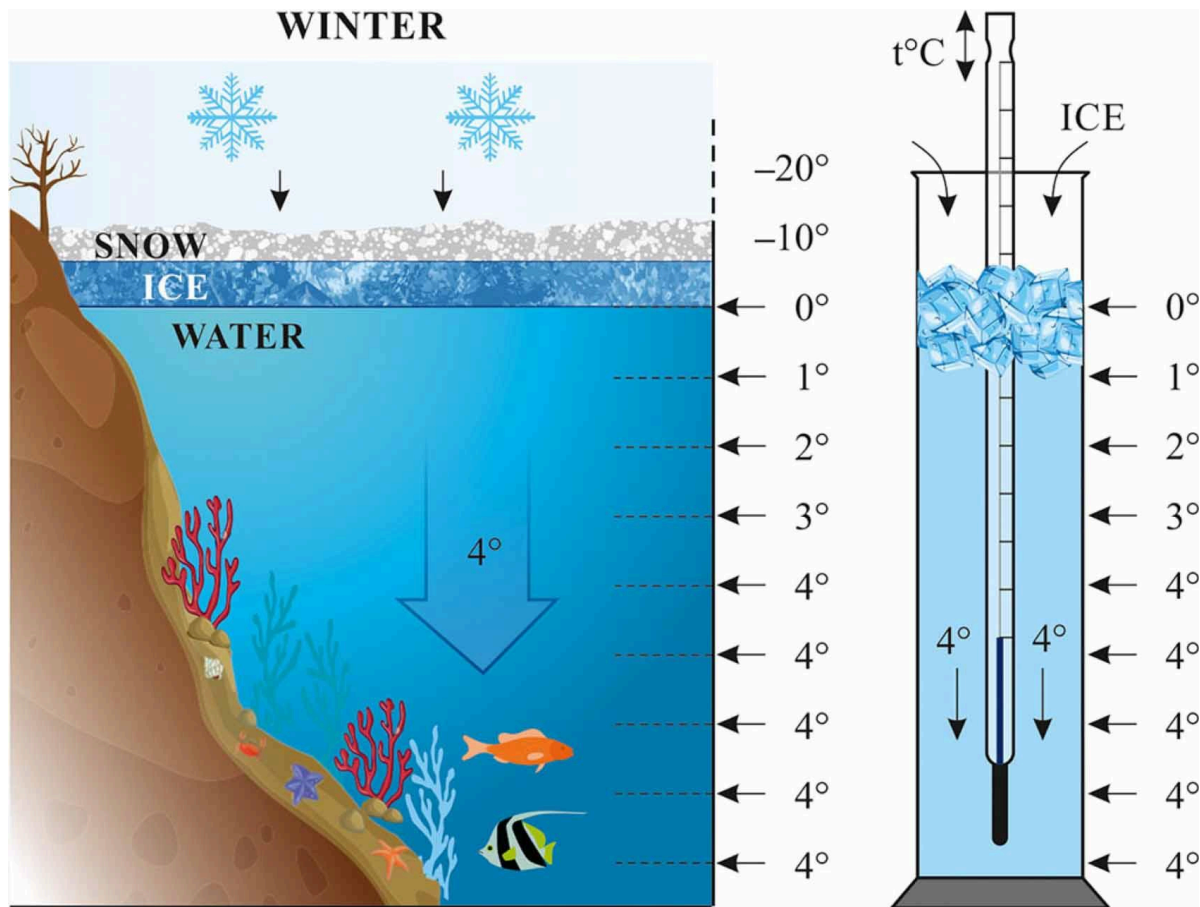
This Venn diagram represents students' understanding of the role of water in plants and animals, highlighting both similarities and differences. The visual structure indicates that students were able to categorise and compare concepts, demonstrating higher-order thinking beyond recall.

4.4 Visual Representations

Visual aids such as diagrams of the water cycle and water distribution were used to support understanding. These visuals helped reduce cognitive load and made concepts more accessible, especially for learners with language challenges.



This artefact was selected as it demonstrates how teacher modelling and visual representation can reduce cognitive load and support conceptual clarity. It also reflects a shift in my practice from explanation-based teaching to facilitated learning through questioning, where the teacher's role is to guide thinking rather than provide answers.



This evidence was selected as it highlights the importance of teacher scaffolding when introducing complex concepts. It shows how demonstration combined with questioning supports conceptual understanding and curiosity, while also emphasising the teacher's role in bridging the gap between observation and scientific explanation.

4.5 Teaching Resources

Teaching resources included presentations, guided questions, and structured worksheets. These resources were designed to scaffold learning and support differentiated questioning. [Powerpoint Presentation Link](#)

4.6 Observation Notes

Observation feedback highlighted the effectiveness of differentiated questioning in increasing student participation and promoting critical thinking. This provided validation of the impact of the approach.

1. Evidence note

During the water cycle lesson, a planned slide could not be projected due to a technical issue. Instead of stopping the lesson, pair discussion was used as an alternative strategy. Students discussed different ways water vapour enters the atmosphere and generated responses such as transpiration and respiration. This observation demonstrates responsive teaching and the ability to maintain cognitive challenge despite disruptions. It also highlights how questioning can support meaningful participation even in unplanned situations.

2. Evidence Note

Observation feedback noted that students who were usually hesitant participated more actively during think pair share discussions. The use of wait time and peer interaction reduced anxiety and encouraged wider participation. This evidence reinforces the importance of questioning strategies that create supportive learning environments.

This artefact was selected because it provides evidence of how differentiated questioning operates not only through prior planning but also through responsive teaching. The open-ended nature of the question encouraged students to generate multiple ideas such as transpiration and respiration, demonstrating higher-order thinking aligned with Bloom's Taxonomy. It also reflects the role of questioning as a form of formative assessment, where student responses were used to gauge understanding and extend thinking. Importantly, this moment reinforced that effective questioning is not dependent on resources but on the teacher's ability to adapt and maintain cognitive challenge. It highlights how inclusion and thinking can still be supported even in unplanned situations, as all students were able to participate through discussion.

4.7 Learner Voice

Student feedback provided valuable insight into learner experiences during the inquiry.

Learner Reflection 1 One thing I liked was that we got time to discuss before answering. It made me feel more confident. This response suggests that structured discussion supported confidence and participation.

Learner Reflection 2 I understood the water cycle better when we explained it ourselves instead of only listening. This highlights the importance of dialogue and active engagement in promoting conceptual understanding.

Learner Reflection 3 The pollution poster activity helped me understand how science connects to real life. This demonstrates that students were able to connect scientific concepts to environmental and social contexts.

Reflection notes documented changes in teaching practice, including the use of scaffolding and structured questioning. These reflections show a shift towards more intentional teaching.

4.8 Reflective Journal

Reflection notes documented changes in my teaching practice throughout the inquiry.

1. I noticed that when I waited longer before expecting answers, more students contributed. Earlier, I often answered my own questions too quickly. 2. I realised that some students struggled not because they did not understand the concept, but because they lacked confidence in expressing scientific vocabulary. 3. The Yamuna River discussion reminded me that students are capable of deeper thinking when given meaningful opportunities to analyse and discuss. These reflections demonstrate how the inquiry shaped my understanding of questioning, participation, and teaching.

5. Reflective Learning Journey

This inquiry has been a process of continuous reflection and professional growth. Initially, questioning functioned mainly as a way to check understanding. Over time, it evolved into a strategy for promoting inclusion, discussion, and thinking. One of the most significant moments occurred during the water cycle lesson when students generated responses such as transpiration and respiration after being given time to think and discuss. This moment challenged my earlier assumptions about student ability and participation. I realised that many students were capable of deeper thinking when provided with appropriate support and opportunities. Another important insight emerged during lessons involving scientific vocabulary. Some students struggled not because they lacked understanding, but because they found it difficult to express ideas confidently. This shifted my understanding of inclusion and highlighted the importance of scaffolding.

This inquiry also changed how I view my role as a teacher. I moved from seeing myself primarily as someone who delivers information towards someone who facilitates thinking and dialogue. This aligns with Schön's concept of reflective practice, where teaching develops through ongoing reflection and adaptation (Schön, 1983). The inquiry also encouraged me to become more comfortable with uncertainty within the classroom. Earlier, I often felt pressure to quickly guide students towards correct answers. However, through this process I began to value moments of confusion and discussion as important parts of learning. Students often developed stronger understanding when they explored misconceptions together rather than immediately receiving teacher explanations.

6. Impact and Professional Learning

This inquiry has had a meaningful impact on both learners and my own teaching practice.

Impact on Learners

Evidence from assessments, student work, and learner feedback indicates:

- Increased participation
- Improved confidence
- Greater conceptual understanding
- Better reasoning and explanation skills

Students demonstrated stronger ability to justify responses and connect science learning to real world contexts. Importantly, the inquiry also revealed that participation is deeply connected to classroom culture. When students felt safe to make mistakes and discuss ideas collaboratively, they became more willing to contribute. This suggests that inclusion is not simply about providing differentiated tasks, but about creating environments where all learners feel intellectually valued.

Impact on Teaching Practice

My questioning shifted from spontaneous questioning towards intentional questioning design. I now plan questions strategically and use them to guide thinking rather than only assess recall. The inquiry strengthened my understanding of formative assessment and responsive teaching. I became more aware of how questioning can either support or limit participation. I also became more aware of the importance of listening carefully to student responses rather than focusing only on correct answers. This inquiry helped me recognise that students' misconceptions provide valuable insight into their thinking and can become opportunities for deeper discussion.

Leadership of Learning: This inquiry reflects my development as a reflective practitioner and emerging Leader of Learning.

Leadership action	Impact
Shared questioning strategies with colleagues	Encouraged collaborative professional dialogue
Reflected on lesson observations	Improved responsiveness during teaching
Adapted questioning based on learner voice	Increased student participation and inclusion

Leadership became less about authority and more about creating environments where learners feel safe to think, question, and participate. This inquiry strengthened my commitment to evidence based reflective practice and collaborative learning. As a developing Leader of Learning, I also recognised the importance of sharing reflective practice with colleagues. Informal professional conversations about questioning strategies encouraged collaborative discussion around inclusion and student participation within the Science department. This process reinforced my understanding that professional growth is strengthened through shared reflection and dialogue.

7. Ethics and AI Declaration

All student work included in this portfolio has been anonymised to protect learner identity and confidentiality. Permissions were obtained where necessary, and all evidence has been used ethically and responsibly.

AI tools were used to support language refinement, organisation, and editing during the preparation of this portfolio. However, all reflections, analysis, classroom evidence, and professional interpretations are based on my own teaching experiences and inquiry process.

8. Conclusion

This inquiry has demonstrated that differentiated questioning can significantly support both inclusion and thinking within the classroom. Through intentional questioning, scaffolding, and collaborative discussion, it is possible to create learning opportunities where all students participate meaningfully and engage more deeply with scientific concepts. The most important learning from this process is that questioning is not simply a teaching strategy, but a way of shaping how students think, participate, and learn. I no longer ask questions only to check understanding. I now ask questions to create thinking, encourage dialogue, and support inclusion.

This inquiry has reinforced my commitment to becoming a reflective practitioner who continuously adapts teaching in response to learner needs and experiences. Moving forward, I will continue refining questioning strategies to support student voice, independent thinking, and inclusive participation across all areas of learning. The inquiry has also strengthened my understanding that effective science teaching is not measured by how much information is delivered, but by the quality of thinking that takes place within the classroom. Through this process I have become more intentional, reflective, and responsive in my practice. Most importantly, I have come to recognise that every learner deserves opportunities to think deeply, contribute meaningfully, and feel valued within the learning process.

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